

Demonstration of Fonts and Alignment Filter for Pandoc

Nandakumar Chandrasekhar

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Introduction

This document showcases the features of the `fonts-and-alignment` Pandoc Lua filter. Each example displays the required Markdown syntax alongside its rendered output, ensuring consistent typography across LaTeX, Typst, and HTML formats.¹

The extension leverages Pandoc's `fenced_divs` and `bracketed_spans` extensions. If these are disabled, the filter will issue a terminal warning and render the elements as raw text instead of applying the styles.

Bracketed Spans and Fenced Divs Invocations

This filter follows Pandoc's standard conventions for container-based styling: use Bracketed Spans for inline text and Fenced Divs for block-level content. For further details on these syntax standards, please refer to [Pandoc User's Guide: Divs and Spans](#).

```
[This is a line of text in extra extra large font.]{.pfa-size-2xl}
```

This is a line of text in extra extra large font.

```
::: {.pfa-size-2xl}
```

This multiline block showcases the filters's capability to apply
→ extra extra large font to an entire section of text by wrapping
→ the content in a Fenced Div.

```
:::
```

This multiline block showcases the filters's capability to apply extra extra large font to an entire

¹HTML output requires the companion `fonts-and-alignment.css` stylesheet.

section of text by wrapping the content in a Fenced Div.

Font Sizing in Bracketed Spans

The extension provides nine predefined sizing hooks to scale text relative to the document's base/normal font size. The *LaTeX Reference* column is provided for descriptive purposes, as these labels are not native to Typst or HTML/CSS.

LaTeX Reference	Syntax	Output
tiny	<code>[Sample]{.pfa-size-3xs}</code>	Sample
scriptsize	<code>[Sample]{.pfa-size-2xs}</code>	Sample
footnotesize	<code>[Sample]{.pfa-size-xs}</code>	Sample
small	<code>[Sample]{.pfa-size-s}</code>	Sample
normal	<code>[Sample]{.pfa-size-normal}</code>	Sample
large	<code>[Sample]{.pfa-size-l}</code>	Sample
Large	<code>[Sample]{.pfa-size-xl}</code>	Sample
LARGE	<code>[Sample]{.pfa-size-2xl}</code>	Sample
huge	<code>[Sample]{.pfa-size-3xl}</code>	Sample

Font Sizing in Fenced Divs

The same sizing classes can also be applied to Fenced Divs, allowing an entire block of text to be rendered at a specific font size.

Extra Small Font Size

```
::: {.pfa-size-xs}
This paragraph renders at the *extra small* font size.
:::
```

This paragraph renders at the *extra small* font size.

Normal Font Size

```
::: {.pfa-size-normal}  
This paragraph renders at the *normal* font size.  
:::
```

This paragraph renders at the *normal* font size.

Large Font Size

```
::: {.pfa-size-l}  
This paragraph renders at the *large* font size.  
:::
```

This paragraph renders at the *large* font size.

Font Weights, Shapes, and Families in Bracketed Spans

The extension provides predefined typographic styles for Bracketed Spans. These styles are mapped to equivalent rendering properties across all formats to ensure consistent output, regardless of the underlying engine.

Typographic Style	Syntax	Output
Normal (weight)	[Sample]{.pfa-weight-normal}	Sample
Medium (weight)	[Sample]{.pfa-weight-medium}	Sample
Bold (weight)	[Sample]{.pfa-weight-bold}	Sample
Emphasis (style)	[Sample]{.pfa-style-emph}	<i>Sample</i>
Upright (style)	[Sample]{.pfa-style-upright}	Sample
Italic (style)	[Sample]{.pfa-style-italic}	<i>Sample</i>
Slanted (style)	[Sample]{.pfa-style-slanted}	<i>Sample</i>
Small Caps (style)	[Sample]{.pfa-style-smallcaps}	SAMPLE
Serif (family)	[Sample]{.pfa-family-serif}	Sample
Sans-Serif (family)	[Sample]{.pfa-family-sans}	Sample
Monospace (family)	[Sample]{.pfa-family-mono}	Sample

Font Weights, Shapes, and Families in Fenced Divs

These same typographic classes can also be applied to Fenced Divs, allowing an entire block of text to adopt a particular visual style.

Bold Weight

```
::: {.pfa-weight-bold}
This paragraph is in bold weight.
:::
```

This paragraph is in bold weight.

Small Caps

```
::: {.pfa-style-smallcaps}
This paragraph is in small caps.
:::
```

THIS PARAGRAPH IS IN SMALL CAPS.

Sans-Serif Family

```
::: {.pfa-family-sans}
This paragraph is in sans-serif.
:::
```

This paragraph is in sans-serif.

Text Casing Transformations

Unlike purely visual styling in HTML/CSS—where `text-transform` only changes how text is rendered—these casing transformations operate at the [Pandoc Abstract Syntax Tree \(AST\)](#) level and modify the actual document content. This ensures that uppercase and lowercase conversions are preserved consistently across all output formats and remain intact when copied into other applications.

Casing Transformations in Bracketed Spans

Casing	Syntax	Output
Uppercase	<code>[Sample]{.pfa-case-upper}</code>	SAMPLE
Lowercase	<code>[SAMPLE]{.pfa-case-lower}</code>	sample

Casing Transformations in Fenced Divs

Lowercase

```

::: {.pfa-case-lower}
THIS PARAGRAPH IS TRANSFORMED TO LOWERCASE.
:::

```

this paragraph is transformed to lowercase.

Uppercase

```

::: {.pfa-case-upper}
this paragraph is transformed to uppercase.
:::

```

THIS PARAGRAPH IS TRANSFORMED TO UPPERCASE.

Colors and Backgrounds

This section describes how to apply colors to text, as well as how to apply background fills, padding, and borders to your content.

Color Fundamentals

The filter normalizes color inputs to ensure they render identically across LaTeX, Typst, and HTML.

Flexible Color Terminology

For solid colors, the filter is case-insensitive and accepts most naming conventions. All examples below resolve to the standard CSS3 `mediumvioletred`:

Naming	Syntax	Output
Lowercase	<code>[Sample]{pfa-color="mediumvioletred"}</code>	Sample

Naming	Syntax	Output
Title Case	<code>[Sample]{pfa-color="Medium Violet Red"}</code>	Sample
Kebab Case	<code>[Sample]{pfa-color="medium-violet-red"}</code>	Sample
Snake Case	<code>[Sample]{pfa-color="medium_violet_red"}</code>	Sample
Pascal Case	<code>[Sample]{pfa-color="MediumVioletRed"}</code>	Sample

Input Types

Input Type	Syntax	Output
CSS3 Named	<code>[Sample]{pfa-color="crimson"}</code>	Sample
Hex Full	<code>[Sample]{pfa-color="#2E8B57"}</code>	Sample
Hex Shorthand ²	<code>[Sample]{pfa-color="#666"}</code>	Sample

Compatibility Note: To ensure cross-platform stability, use standard CSS3 named colors or hex codes. Avoid LaTeX-specific numbered variants (e.g., `LightBlue3`), as these are not supported in web browsers or CSS.

Color Mixing

The extension supports LaTeX's `xcolor` percentage-mixing syntax for tints, shades, and blends. This is the standardized syntax because it maps reliably across Typst, LaTeX, and HTML (which leverages the native CSS `color-mix()` function).

Supported Mixing Methods

- **Tinting (blending with white):** Use the pattern `BaseColor!Percentage`. For example, `Maroon!40` yields 40% Maroon and 60% white.

²Three-digit shorthand expands by duplicating each single hexadecimal digit per RGB channel (e.g., `#666` expands to `#666666`).

- **Shading (blending with black):** Use the pattern `BaseColor!Percentage!black`. For example, `MediumVioletRed!80!black` yields 80% MediumVioletRed and 20% black.
- **Two-Color Mix:** Use the pattern `BaseColor!Percentage!MixColor`. For example, `RoyalBlue!50!ForestGreen` yields a 50/50 mix of both colors.

Syntax Examples

Mixing Type	Syntax	Output
Tint (40% Base)	<code>[Sample]{pfa-color="Maroon!40"}</code>	Sample
Shade (80% Base)	<code>[Sample]{pfa-color="MediumVioletRed!80!black"}</code>	Sample
Mix (50/50)	<code>[Sample]{pfa-color="RoyalBlue!50!ForestGreen"}</code>	Sample

Important Compatibility Rules

- **Strictly Use LaTeX Syntax:** Always use the `BaseColor!Percentage` pattern. This is the only syntax guaranteed to translate correctly across all three output formats.
- **Avoid Typst Native Syntax:** If you use native Typst color syntax (e.g., `color.mix()`), the filter will pass it directly to Typst. It will not be translated for LaTeX or HTML.
- **Binary Mixing Only:** You may blend a maximum of two colors. Multi-color mixing (e.g., `Red!30!Blue!30!Green`) is not supported.
- **Casing Matters:** Because mixing strings are passed directly to LaTeX, casing must be exact:
 - Use lowercase for **Core LaTeX colors** (e.g., `black`, `red`).
 - Use PascalCase for **CSS3 named colors** (e.g., `RoyalBlue`, `MediumVioletRed`).

Foreground Styling

Use the `pfa-color` attribute to change the color of the text itself. It can be applied to inline spans or multiline blocks.

```
This is normal text, but [Sample]{pfa-color="tomato"} overrides the  
↪ default text color.
```

This is normal text, but **Sample** overrides the default text color.

Backgrounds and Borders

By combining background fills, padding, borders, and corner radii, you can highlight and emphasize text. These attributes can be applied to both inline bracketed spans and multiline fenced divs.

The Core Attributes

- **pfa-bg-color**: Defines the background fill color of the container.
- **pfa-padding**: Controls internal spacing using standard CSS shorthands. Only **pt**, **em**, and **ex** are supported to guarantee cross-platform stability.
 - One value (**1em**): Applied to all four sides evenly.
 - Two values (**1em 0.5em**): Top/Bottom, then Left/Right.
 - Three values (**1em 0.5em 2em**): Top, Left/Right, Bottom.
 - Four values (**1em 0.5em 2em 0.2em**): Top, Right, Bottom, Left.
- **pfa-bg-border-width**: Defines the thickness of the border around all four sides. We strongly recommend using **pt** to ensure crisp print rendering. If this is omitted, the border defaults to **1pt**.
- **pfa-bg-border-color**: Defines the color of the border.
- **pfa-bg-radius**: Rounds all four corners of the container uniformly. We recommend using **em** so the curve scales naturally with your text size.

Example 1: Inline Spans

```
This is a [Sample]{pfa-bg-color="aliceblue" pfa-color="navy"  
↪ pfa-padding="0.2em 0.6em" pfa-bg-border-width="0.5pt"  
↪ pfa-bg-border-color="steelblue" pfa-bg-radius="1em"} of an inline  
↪ badge.
```

This is a  of an inline badge.

Example 2: Multiline Blocks

```
::: {pfa-bg-color="ghostwhite" pfa-padding="1.5em"  
  ↪ pfa-bg-border-width="1.5pt" pfa-bg-border-color="maroon"  
  ↪ pfa-bg-radius="0.4em"}  
**Important Note:** This block utilizes the full suite of styling  
  ↪ attributes to create a container with a light background, a  
  ↪ maroon border, and rounded corners.  
:::
```

Important Note: This block utilizes the full suite of styling attributes to create a container with a light background, a maroon border, and rounded corners.

Typst Named Palette

You can access Typst's native predefined colors by prefixing the color name with `typst`.

- **Syntax:** `[Sample]{pfa-bg-color="typstmaroon" pfa-color="white" pfa-padding="0.2em"}`
- **Output:** Sample

Text Alignment within Fenced Divs

The `pfa-align-*` classes control text alignment within a Fenced Div while explicitly preserving hard line breaks (`\`). These utilities map directly to native LaTeX commands (`\raggedright`, `\centering`, and `\raggedleft`) in PDF output and equivalent CSS behaviors in HTML, making them ideal for formatting poetry, lyrics, or multi-line blocks.

Left-aligned Text

```
::: {.pfa-align-left}  
This block of text is *left-aligned*.  
:::
```

This block of text is *left-aligned*.

Center-aligned Text

```
::: {.pfa-align-center}
This block of text is *center-aligned*.
:::
```

This block of text is *center-aligned*.

Right-aligned Text

```
::: {.pfa-align-right}
This block of text is *right-aligned*.
:::
```

This block of text is *right-aligned*.

Line Break Preservation

```
::: {.pfa-align-center}
This block of text\
is *center-aligned*\
while preserving explicit line breaks.
:::
```

This block of text
is *center-aligned*
while preserving explicit line breaks.

Combining Multiple Classes

Multiple typographic utilities can be combined within the same element. Font, size, color, alignment, and decoration classes are designed to compose independently and can be applied together to both Bracketed Spans and Fenced Divs.

Class names are space-separated within `{ }` following Pandoc attribute syntax. In addition to classes, key-value attributes such as `pfa-color` may be included alongside class definitions.

Bracketed Span Composition

Style	Syntax	Output
Bold Sans-Serif	<code>[Sample]{.pfa-wight-bold .pfa-family-sans .pfa-size-l pfa-color="red"}</code>	Sample
Italic Monospace	<code>[Sample]{.pfa-style-italic .pfa-family-mono .pfa-size-s}</code>	<i>Sample</i>
Small Caps, Colored	<code>[Sample]{.pfa-style-smallcaps pfa-color="forestgreen"}</code>	SAMPLE

Fenced Div Composition

Multiple utilities can also be applied to Fenced Divs to control alignment, typography, and color simultaneously. The resulting block inherits all specified styles while preserving Pandoc's standard attribute behavior.

```
::: {.pfa-align-center .pfa-family-sans .pfa-weight-bold .pfa-size-l  
↪ pfa-color="midnightblue"}  
A centered, bold, sans-serif, large, midnight-blue Fenced Div  
↪ demonstrating multiple combined utilities from the filter.  
:::
```

**A centered, bold, sans-serif, large, midnight-blue Fenced Div
demonstrating multiple combined utilities from the filter.**

Legacy Aliases (Deprecated)

Warning: Legacy aliases are deprecated. They are retained strictly for backward compatibility and will be entirely removed in the next major release. New documents should use the `pfa-*` namespaces going forward.

Font Sizing Aliases

Legacy Class	Modern Class
<code>xsmall</code>	<code>pfa-text-xs</code>
<code>small</code>	<code>pfa-text-s</code>
<code>normal</code>	<code>pfa-text-normal</code>
<code>large</code>	<code>pfa-text-l</code>

Legacy Class	Modern Class
xlarge	pfa-text-xl
xxlarge	pfa-text-2xl
huge	pfa-text-3xl

Font Weight, Shape, and Family Aliases

Legacy Class	Shorthand	Modern Class
bold	bf	pfa-font-bold
emphasis	em	pfa-font-emphasis
italic	it	pfa-font-italic
medium	md	pfa-font-medium
monospace	tt	pfa-font-mono
normalfont	nf	pfa-font-normal
sans	sf	pfa-font-sans
serif	rm	pfa-font-serif
slanted	sl	pfa-font-slanted
smallcaps	sc	pfa-font-smallcaps
upright	up	pfa-font-upright

Alignment Aliases

Legacy Class	Modern Class
centering	pfa-align-center
raggedleft	pfa-align-right
raggedright	pfa-align-left

Text Decoration Aliases

Legacy Class	Shorthand	Modern Class
<code>uline</code>	<code>u</code>	<code>pfa-text-underline</code>
<code>uuline</code>	<code>uu</code>	<code>pfa-text-underline-double</code>
<code>dashuline</code>	<code>dau</code>	<code>pfa-text-underline-dashed</code>
<code>dotuline</code>	<code>dou</code>	<code>pfa-text-underline-dotted</code>
<code>uwave</code>	<code>uw</code>	<code>pfa-text-underline-wave</code>
<code>sout</code>	<code>so</code>	<code>pfa-text-strikeout</code>

Removed Classes

The following legacy classes were removed in version 2.0.0 and are no longer recognized by the filter. Existing documents should be updated to use the replacement classes shown below.

Removed Class	Shorthand	Removed In	Replacement
<code>center</code>	—	2.0.0	<code>pfa-align-center</code>
<code>flushleft</code>	—	2.0.0	<code>pfa-align-left</code>
<code>flushright</code>	—	2.0.0	<code>pfa-align-right</code>
<code>xout</code>	<code>xo</code>	2.0.0	<code>pfa-text-strikeout</code>